

Final Report: Analysis of COVID-19 Government Policies & Human Mobility

DSA210 Individual Project — Fall 2025-2026

Abstract

This project analyzes how COVID-19 government policies affected human mobility across 11 countries (Feb 2020 – Jan 2021) using Oxford Government Response Tracker and Google Mobility Reports. Statistical tests and machine learning were used to validate policy-mobility relationships. Results supported all four hypotheses: stricter policies significantly reduced mobility, though this effect weakened in later phases. Additionally, machine learning models showed that government mandates are not the sole driver of behavior ($R^2 \approx 0.24$), with cultural and economic context playing a major role.

Motivation

I wanted to determine if strict government laws are enough to control human behavior during a crisis, or if people stop complying over time due to economic necessity or fatigue. The pandemic created an unprecedented global-scale natural experiment to explore this question.

Data Description and Processing

Sources & Variables: I analyzed data from two open-source global repositories:

1. **Oxford COVID-19 Government Response Tracker:** Provided the Stringency Index (0–100) and specific policy indicators (e.g., C2: Workplace Closing).
2. **Google Community Mobility Reports:** Provided daily movement changes relative to a Jan 2020 baseline across six categories (e.g., Workplaces, Retail).

Data Description: The dataset tracks two main things. **Policy data** is measured using a “Stringency Index” (0 – 100) and simple rankings for specific rules (e.g., 0 = no rule, 3 =strict enforcement).

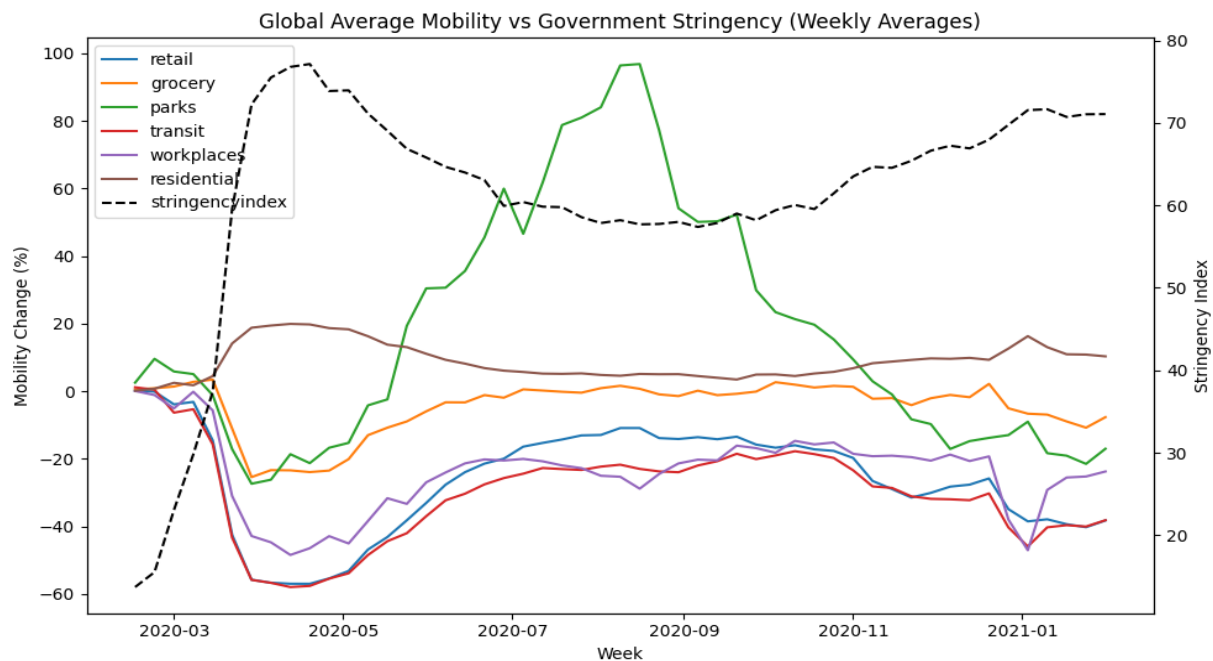
Mobility Data measures the percentage change in activity compared to normal time. For example, a value of -40 in *Workplaces* means activity was %40 lower than the pre-pandemic baseline.

Data Processing: I merged these datasets by country and date to create a unified timeline. To ensure data quality, I removed columns with excessive missing values (e.g. confirmedcases, %70 missing values) and I grouped the daily data into weekly averages to get rid of random spikes and make the trends easier to see.

Exploratory Data Analysis

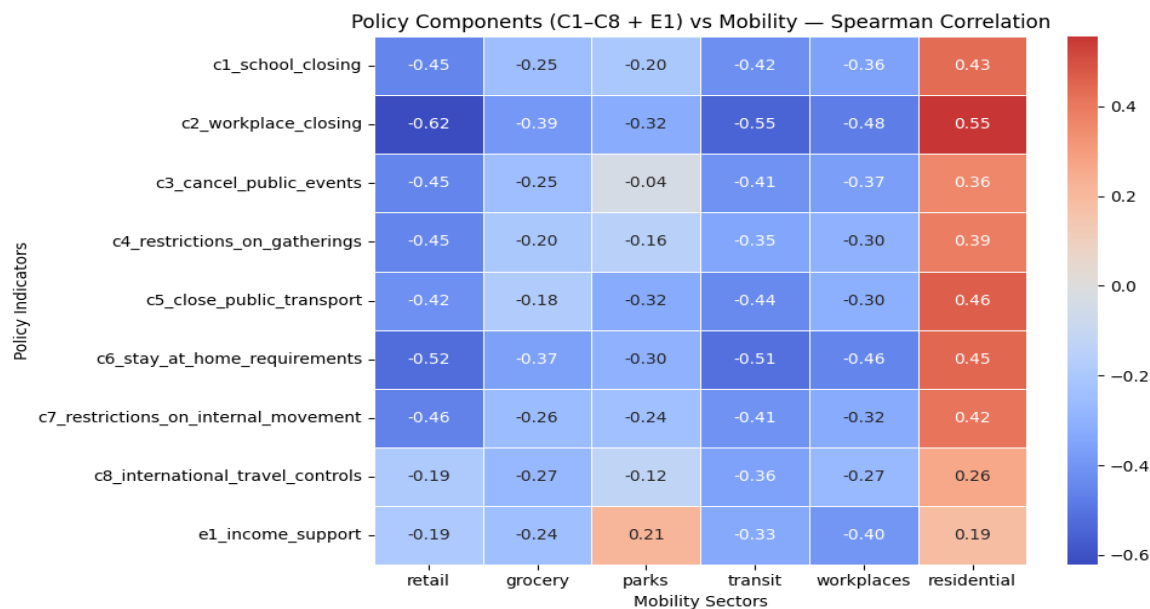
Before testing hypotheses, I explored the data to understand patterns and relationships. I visualized global mobility trends over time, examined correlations between policy indicators and mobility sectors, checked whether mobility sectors move together, and compared how different countries responded to similar policies.

Global Mobility Trends



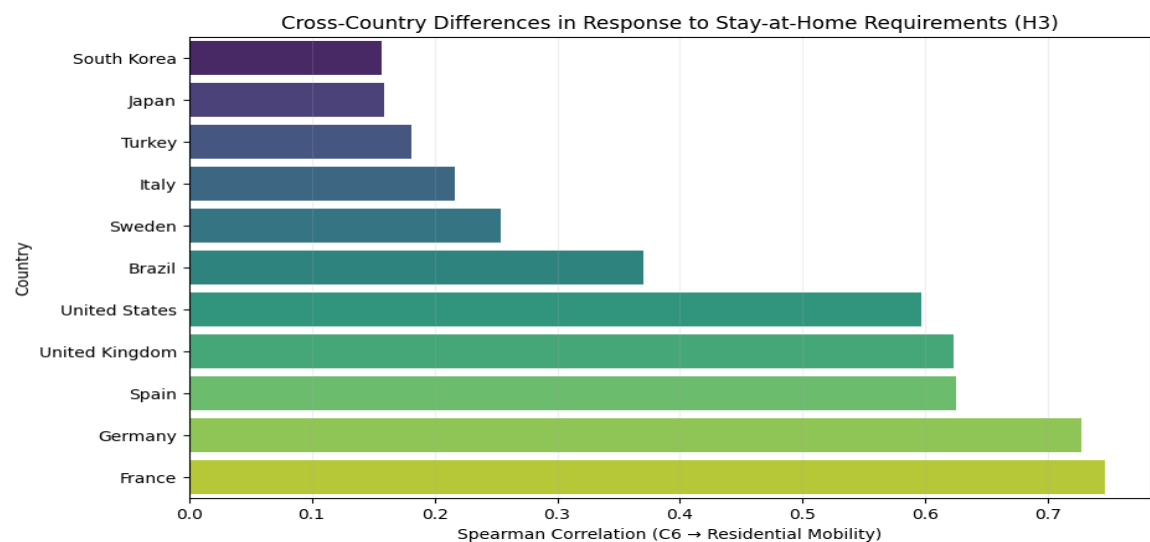
Interpretation: Mobility dropped sharply in March 2020 when stringency spiked. Residential increased as people stayed home, parks spiked in summer. Later, mobility started recovering even while stringency remained high, possibly because people stopped complying with restrictions.

Policy-Mobility Correlations



Interpretation: C2 (workplace closing) and C6 (stay-at-home) show the strongest negative correlations with mobility, mostly with retail, transit, and workplaces. However residential mobility is the opposite. It increases when policies are stricter. Parks has the weakest relationship with any policy indicator.

Cross-Country Variation



Interpretation: France, Germany, and Spain showed the strongest response to stay-at-home requirements ($\rho \approx 0.62\text{--}0.75$), while South Korea and Japan had the weakest ($\rho \approx 0.25\text{--}0.37$). Same policy, different responses — cultural and compliance factors likely play a role.

Hypotheses & Findings

H1 — Policy-Mobility Association

Stricter government policies are associated with reduced mobility.

Test: Kendall's τ correlation

	policy	mobility	kendall_tau	p_value
0	c2_workplace_closing	workplaces	-0.380	4.39e-208
1	c2_workplace_closing	retail	-0.503	0
2	c2_workplace_closing	transit	-0.441	4.66e-280
3	c2_workplace_closing	residential	0.450	1.5e-283
4	c6_stay_at_home_requirements	workplaces	-0.364	4.25e-184
5	c6_stay_at_home_requirements	retail	-0.424	5.76e-250
6	c6_stay_at_home_requirements	transit	-0.414	6.25e-238
7	c6_stay_at_home_requirements	residential	0.369	1.07e-184

Supported. All p-values < 0.05, reject H0. Policies and mobility are significantly associated — negative for out-of-home sectors, positive for residential.

H2 — Sector Spillover

Mobility sectors move together rather than independently.

Test: Spearman correlation between sectors

mobility_1	mobility_2	spearman_rho	p_value
workplaces	retail	0.509	4.26e-254
workplaces	transit	0.786	0
workplaces	grocery	0.534	8.34e-285
workplaces	parks	0.144	1.81e-19

Supported. Out-of-home sectors (workplace, retail, transit) are strongly positively correlated. Residential shows negative correlation when people go out less, they stay home more.

H3 — Cross-Country Variation

Different countries respond differently to the same policies.

	country	spearman_rho	p_value	n_obs
1	France	0.746	9.97e-64	352
2	Germany	0.727	3.63e-59	352
6	Spain	0.625	1.43e-39	352
9	United Kingdom	0.623	3.38e-39	352
10	United States	0.596	2.79e-35	352
0	Brazil	0.370	7.13e-13	352
7	Sweden	0.254	1.42e-06	352
3	Italy	0.217	4.19e-05	352
8	Turkey	0.181	0.000629	352
4	Japan	0.159	0.00276	352
5	South Korea	0.157	0.00312	352

Supported. Same policy, different responses. European countries showed stronger compliance than Asian countries in this metric.

H4 — Temporal Weakening

Policy effects weaken over time as people stop complying.

Phase	Policy	Mobility	Spearman ρ
Phase 1 (first lockdown)	c5_close_public_transport	transit	-0.597
Phase 2 (re-opening)	c5_close_public_transport	transit	-0.367
Phase 3 (winter wave)	c5_close_public_transport	transit	-0.121
Phase 1 (first lockdown)	c7_internal_movement	retail	-0.601
Phase 2 (re-opening)	c7_internal_movement	retail	-0.390
Phase 3 (winter wave)	c7_internal_movement	retail	-0.093

Supported. Correlation strength drops from Phase 1 to Phase 3. People stopped complying with restrictions over time.

Machine Learning

ML was used to evaluate whether mobility can be predicted from policy indicators, extending beyond statistical associations.

Supervised Learning

- *Regression* — Predict workplace mobility level (Linear Regression, Lasso, Random Forest)
- *Classification* — Detect large mobility drops (Logistic Regression, Random Forest)

Unsupervised Learning

- *Clustering* — Identify country response patterns (K-Means)

Regression Results:

	Model	RMSE	MAE	R2
0	Linear Regression	16.224722	12.163996	0.240354
1	Lasso Regression	16.224693	12.163013	0.240357
2	Random Forest Regressor	17.855254	12.781523	0.079999

Best: Linear/Lasso ($R^2 \approx 0.24$). Relationship is mostly linear.

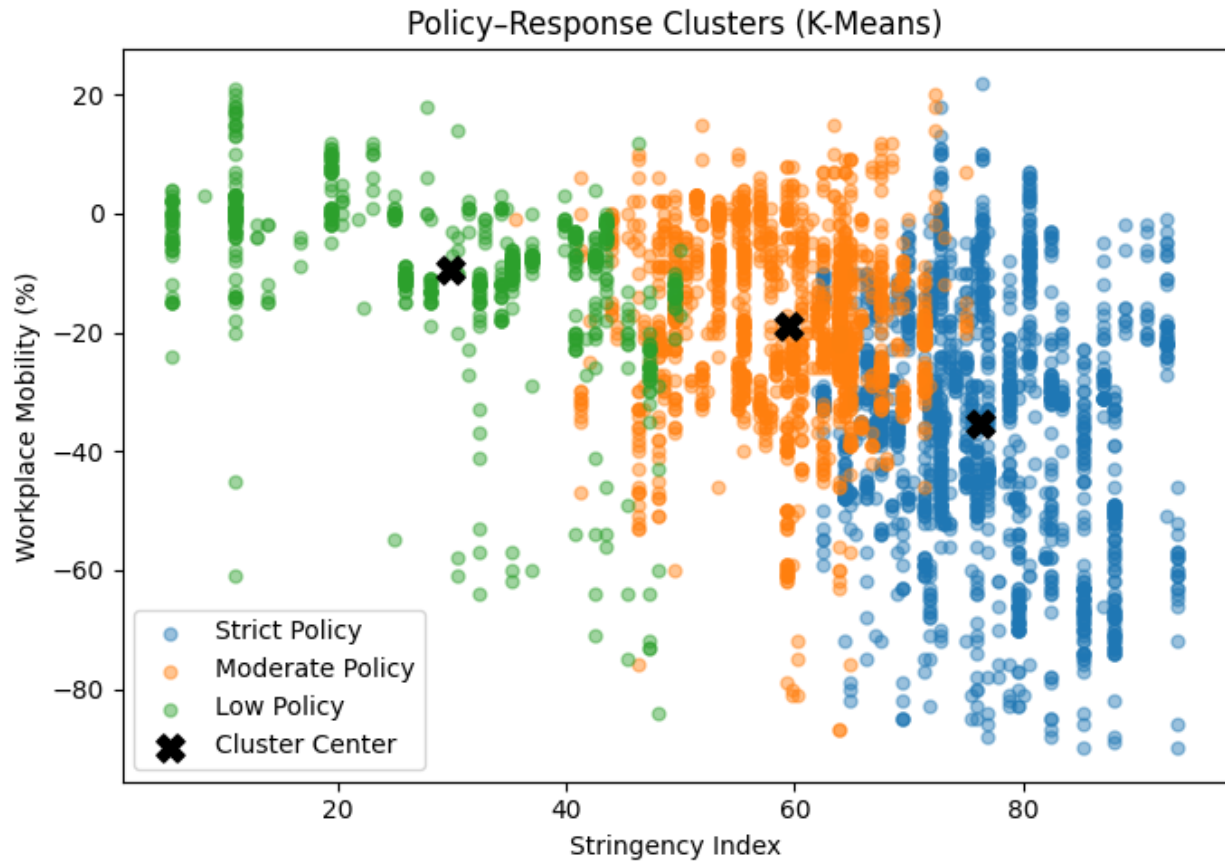
Classification Results:

	Model	Accuracy	Precision	Recall	F1
0	Logistic Regression	0.723871	0.744376	0.803532	0.772824
1	Random Forest Classifier	0.709677	0.740506	0.774834	0.757282

Best: Logistic Regression (F1 = 0.77). Can detect large mobility drops fairly well.

Clustering Results:

K-Means identified 3 clusters: Strict Policy, Moderate Policy, Low Policy.



Interpretation: Low Policy cluster (green) showed large mobility drops even with lower stringency, people stayed home voluntarily, not because of strict rules.

ML Conclusion:

Policies matter, but they only explain ~24% of mobility changes. Human behavior is more complex, cultural norms, fear, and economic factors play a major role.

Conclusion

All four hypotheses were supported. Policies significantly reduced mobility, sectors moved together, responses varied among countries, and effects weakened over time as people stopped complying.

Machine learning showed that policies explain only 24% of mobility changes.

This shows that policies matter, but people are more complex than what models can fully capture.

Limitations

One of the main limitations of this study is that it relies on observational data , it can prove correlation but not causation. Additionally, the country count was limited to 11 due to data quality issues in other countries. Policy indicators also reflect what was announced, not how strictly it was enforced.

Future Work

As future work, if variables such as economic indicators or cultural context are added to the study with a higher count of countries, the findings may be able to predict mobility changes more accurately.